RAJALAKSHMI ENGINEERING COLLEGE

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M2

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| **Started on** | Monday, 31 August 2026, 8:04 PM |
| **State** | Finished |
| **Completed on** | Monday, 31 August 2026, 8:04 PM |
| **Time taken** | 30 secs |
| **Marks** | 1.00/1.00 |
| **Grade** | **10.00** out of 10.00 (**100**%) |

**Question 1** | Correct Mark 1.00 out of 1.00

Given an array nums of size n, return *the majority element*.

The majority element is the element that appears more than $\lfloor n / 2 \rfloor$ times. You may assume that the majority element always exists in the array.

**Example 1:**

```
Input: nums = [3,2,3]
Output: 3
```

**Example 2:**

```
Input: nums = [2,2,1,1,1,2,2]
Output: 2
```

**Constraints:**

- $n == nums.length$
- $1 <= n <= 5 * 10^4$
- $-2^{31} <= nums[i] <= 2^{31} - 1$

**For example:**

| Input | Result |
|---|---|
| 3<br>3 2 3 | 3 |
| 7<br>2 2 1 1 1 2 2 | 2 |

**Answer:** (penalty regime: 0 %)

```c
#include <stdio.h>

int majorityElement(int nums[], int low, int high) {
    if (low == high)
        return nums[low];

    int mid = (low + high) / 2;

    int left = majorityElement(nums, low, mid);
    int right = majorityElement(nums, mid + 1, high);

    if (left == right)
        return left;

    int leftCount = 0, rightCount = 0;

    for (int i = low; i <= high; i++) {
        if (nums[i] == left)
            leftCount++;
        if (nums[i] == right)
            rightCount++;
    }

    return leftCount > rightCount ? left : right;
}
```

```
int main() {
    int n;
    scanf("%d", &n);

    int nums[n];

    for (int i = 0; i < n; i++)
        scanf("%d", &nums[i]);

    printf("%d\n", majorityElement(nums, 0, n - 1));

    return 0;
}
```

| | Input | Expected | Got | |
|---|---|---|---|---|
| | 3<br>3 2 3 | 3 | 3 | |

Passed all tests!

Correct

Marks for this submission: 1.00/1.00.

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